

Academic and Research Integrity

Introduction

Academic and research integrity means being honest, fair, and responsible in study, experiments, testing, data collection, and publication. In engineering, this is very important because wrong research or false data can lead to unsafe products, failed designs, public harm, and loss of trust.

According to Fleddermann, research ethics has two major issues: **honesty in approaching the research problem** and **honesty in reporting the results**. A researcher must remain objective, avoid preconceived conclusions, and report data accurately.

1. Honesty in Approaching Research

A researcher should begin research with an open and objective mind. They should not decide the result before doing the experiment. The purpose of research is to discover truth, not to prove what the researcher already wants to believe.

If evidence does not support the original hypothesis, the researcher should be ready to change the hypothesis. Fleddermann says this objective attitude is necessary for producing useful research and testing results.

2. Honesty in Reporting Results

After an experiment is completed, the results must be reported accurately. A researcher must not exaggerate, hide, change, or “massage” the data to get the desired outcome.

Fleddermann explains that academic pressure, desire for fame, and pressure to be first can tempt researchers to overstate results or falsify data. In industry, pressure to bring a product to market can also tempt test engineers to adjust data dishonestly.

3. Difference Between Error and Fraud

Every incorrect result is not fraud. Sometimes a researcher may publish a result that later turns out to be wrong because of measurement error, weak method, or lack of knowledge.

This becomes unethical when the researcher intentionally deceives others or refuses to correct the result after discovering the mistake. Fleddermann says great care must be taken before results are reported, and corrections should be made when needed.

4. Giving Proper Credit

Research is usually not done by one person alone. Many people may help through ideas, guidance, experiments, apparatus design, data collection, or interpretation.

Therefore, all contributors must be acknowledged properly. Not giving credit is dishonest. Fleddermann explains that professional codes require engineers to recognize the contributions of others and present information in an objective and truthful manner.

5. Academic Integrity

Academic integrity means honesty in university work such as assignments, reports, exams, projects, and research papers. A student must not cheat, copy, plagiarize, fabricate data, or submit someone else's work as their own.

For engineering students, academic dishonesty is serious because it develops habits that may later become professional dishonesty. A student who fakes lab data today may become an engineer who falsifies safety test results tomorrow.

6. Research Fraud

Research fraud means intentional dishonesty in research work. It damages science because other researchers, industries, and the public may rely on false results.

Common forms of research fraud include:

A. Fabrication / Forging

This means inventing data or results that were never actually obtained.

Example:

Alice writes experimental readings even though she never performed the experiment.

B. Falsification / Cooking

This means changing data, methods, or results to make them support the desired conclusion.

Example:

Alice changes failed readings so that the material appears stronger than it really is.

C. Trimming

This means smoothing or adjusting data slightly so that the results look cleaner and more accurate than they actually are.

Example:

Removing small differences in readings to make a graph look perfect.

D. Plagiarism / Wrong Credit

This means using someone else's words, ideas, data, code, or research without proper credit.

Example:

Copying another researcher's literature review or experimental method without citation.

7. Conflict of Interest and Bias

Research should be unbiased, but conflict of interest can create pressure. For example, if a company funds research, the researcher may feel pressure to produce results that favor that company.

Fleddermann discusses ghostwriting and pharmaceutical research as examples where funding and hidden contributions may create ethical concerns. Research papers should clearly disclose funding, conflicts, and real contributors.

8. Pathological Science

Pathological science means a researcher reaches false conclusions because of self-deception, wishful thinking, or strong belief in their own theory. It does not always involve intentional fraud, but it still damages research quality.

Fleddermann says self-deception is a major problem in research. Researchers must be careful, objective, and open to criticism because the goal of research is not fame or publicity, but discovery of new knowledge.

Conclusion

Academic and research integrity is essential for trustworthy engineering. Researchers and students must be honest in experiments, data, reports, authorship, and publication.

A good engineer must not fabricate, falsify, hide, or exaggerate data. They must give proper credit, disclose conflicts of interest, correct mistakes, and remain objective. Integrity protects public trust, scientific knowledge, and professional responsibility.

How to Apply in Case Studies

In a case study, first identify the dishonest act. It may be fabricated data, changed results, plagiarism, hidden conflict of interest, or failure to give credit.

Then explain which responsibility is violated: truthfulness, honesty, objectivity, proper credit, or public trust.

After that, mention the harm. False research can mislead other researchers, damage the university, waste money, harm public safety, and destroy professional reputation.

Finally, give the ethical decision: the researcher should stop the publication, report the truth, correct the data, give proper credit, and repeat the experiment honestly if needed.

Case Study: Alice Fabricating Data

Alice is part of an engineering research team working on a new energy-efficient material. She manipulates experimental data so that it matches the desired outcome.

This is a clear violation of research integrity. Alice has broken truthfulness responsibility because she is not reporting what actually happened in the experiment. She is giving false information to her team, journal, university, and future users of the research.

Her action is unethical because fabricated data can mislead other researchers and may lead to unsafe engineering applications. The team should not publish the paper until the data is checked, corrected, and repeated honestly. Alice's misconduct should also be reported through proper academic or institutional procedure.

Memorization Checklist

Remember academic and research integrity with:

O-R-C-F

O — Objectivity: Start research with an open mind.

R — Reporting: Report results accurately and truthfully.

C — Credit: Give proper credit to all contributors.

F — Fraud: Avoid fabrication, falsification, trimming, and plagiarism.

5-Mark Exam Answer

Academic and research integrity means honesty, fairness, and responsibility in study, experiments, testing, data collection, and publication. According to Fleddermann, research ethics has two major issues: honesty in approaching the research problem and honesty in reporting the results. A researcher must remain objective, avoid preconceived conclusions, and change the hypothesis if evidence requires it.

Research results must be reported accurately. A researcher should not exaggerate, hide, falsify, or “massage” data to get the desired outcome. Academic pressure, fame, publication, funding, or product deadlines may tempt researchers to manipulate results, but this is unethical. There is also a difference between honest error and fraud. A wrong result is not always unethical, but intentional deception or refusal to correct an error is unethical.

Integrity also requires giving proper credit to all contributors. Not acknowledging someone’s ideas, data, writing, or experimental help is dishonest. Research fraud includes fabrication, falsification, trimming, and plagiarism. Therefore, students and engineers must be truthful, objective, fair, and careful because false research can damage public trust, waste resources, and create unsafe engineering decisions.